

BTEC Assignment Brief

Qualification	Pearson BTEC Level 5 Higher Nationals in Digital Technologies
Unit number and title	Unit 37: Application Program Interfaces
Learning aim(s)	By the end of this unit students will be able to: LO1 Examine what an API is, the need for APIs and types of APIs LO2 Apply the knowledge of API research to design an application that incorporates relevant APIs for a given scenario or a substantial student chosen application LO3 Implement an application in a suitable development environment LO4 Document the testing of the application, review and reflect on the APIs used
Assignment title	API Implementation, Testing and Security Evaluation
Assessor	Javokhir Fayzullayev
Issue date	
Hand in deadline	

Context	Students will continue from their approved API design solution and move into implementation and testing. Students are required to implement the proposed API solution within an existing application framework and perform structured and functional testing to assess performance and security.
Task	Students must: • Build upon an existing application framework to implement the designed API solution • Utilise appropriate development tools, environments and best practices • Refine and enhance the application using multiple APIs where relevant • Conduct structural and functional testing, documenting the results • Identify security vulnerabilities and risks • Evaluate the performance and security of APIs used
Evidence Required	Students must submit: • Working application including API integration • Source code / repository link • Testing documentation: • structural testing • functional testing • results • Security findings / vulnerability report • Evaluation of the APIs used

Learning Outcomes and Assessment Criteria

Pass		Merit	Distinction
LO1 Examine what an API is, the need for APIs and types of APIs			D1 Evaluate potential security issues surrounding APIs. D2 Create a design for a chosen substantial application that will utilise a range of APIs, justifying choices.
P1 Examine the relationship between an API and a software development kit (SDK)	M1 Assess a range of APIs for a particular platform that covers a range of uses..		
LO2 Apply the knowledge of API research to design an application that incorporates relevant APIs for a given scenario or a substantial student chosen application			
P2 Analyse an existing application that could be extended with a suitable API. M2 Design an application that will utilise an API for a given purpose.	M2 Design an application that will utilise an API for a given purpose.		
LO3 Implement an application in a suitable development environment			D3 Construct an application utilising multiple APIs, following the designs in LO2. D4 Critically evaluate the APIs used within your application. Provide a data security report of your application.
P3 Build on an existing application framework to implement an API.	M3 Develop an application that utilises an API.		
LO4 Document the testing of the application, review and reflect on the APIs used			
P4 Design and complete a 'white box' test of the application, recording the results.	M4 Conduct 'black box' tests of your application, recording the results. M5 Update the application accordingly with the results		